

Data Analysis of E-Commerce Data Using Python and AI Assistance

Project Overview

This capstone project focuses on analyzing e-commerce sales data using Python, data analytics libraries, and AI-assisted prompt engineering techniques. The project aims to perform data cleaning, exploratory data analysis (EDA), business performance evaluation, and dashboard reporting to derive meaningful business insights from transactional sales data. The analysis includes sales trends, category-level revenue contribution, profitability analysis, and target versus actual sales comparison.

Dataset Summary

Dataset / Area	Description	Key Columns Used
List of Orders	Contains customer order-level information.	Order ID, Order Date, State, City
Order Details	Contains transactional sales details.	Category, Amount, Profit, Quantity
Sales Target	Contains monthly category-level sales targets.	Month, Category, Target
Data Cleaning	Performed preprocessing and validation checks.	Missing values, duplicates, formatting
EDA & Reporting	Business analysis and visualization reporting.	Sales trends, profitability, target analysis

Sales Performance Dashboard

Monthly Sales Trend

